

Page replacement:

FIFO

Input:

```
GNU nano 7.2                                fifo.c *
#include <stdio.h>
int main(){
    int pages[] = {1,2,3,1,4,5};
    int frames[3] = {-1,-1,-1};
    int i,j,k = 0, fault = 0, found;
    for (i=0;i<6;i++){
        found = 0;
        for (j=0;j<3;j++){
            if (frames[j] == pages[i]){
                found = 1;
                break;
            }
        }
        if (!found){
            frames[k] = pages[i];
            k = (k+1)%3;
            fault++;
        }
        printf("Page %d -> ",pages[i]);
        for (j=0;j<3;j++){
            printf("%d ",frames[j]);
        }
        printf("\n");
    }
    printf("Page faults = %d\n",fault);
    return 0;
}
```

Output:

```
jannani@jannani:~/pr$ mkdir fifo
jannani@jannani:~/pr$ cd fifo
jannani@jannani:~/pr/fifo$ nano fifo.c
jannani@jannani:~/pr/fifo$ gcc fifo.c
jannani@jannani:~/pr/fifo$ ./a.out
Page 1 -> 1 -1 -1
Page 2 -> 1 2 -1
Page 3 -> 1 2 3
Page 1 -> 1 2 3
Page 4 -> 4 2 3
Page 5 -> 4 5 3
Page faults = 5
jannani@jannani:~/pr/fifo$
```